

Experiment 5

Calculation of Group Velocity, Phase Velocity and Phase Constant of EM Waves in Rectangular Waveguides

Test Case 1

A rectangular waveguide has dimensions $a = 4.0$ cm, $b = 2.0$ cm and operating frequency $f = 8$ GHz. (1) Calculate the cutoff frequency for the TE₁₀ mode. (2) Predict the phase and group velocities for the TE₁₀ mode. (3) If the width is reduced to $a = 3.0$ cm, determine the effect on cutoff frequency and propagation characteristics.

Code -

```
clc; clear; close;  
V=3*10^8;  
a=0.04; b=0.02; f=8*10^9;  
m=1; n=0;  
fc=(V/2)*sqrt((m/a)^2+(n/b)^2);  
disp(fc*10^-9,'Cutoff frequency (GHz) =');  
vp=V/sqrt(1-(fc/f)^2);  
disp(vp*10^-6,'Phase velocity (Mm/sec) =');  
vg=V*V/vp;  
disp(vg*10^-6,'Group velocity (Mm/sec) =');
```

Output:

Original dimensions: $a = 4.0$ cm, $b = 2.0$ cm, $f = 8$ GHz, mode TE₁₀

Parameter	Simulation Result	Theoretical Result
Cutoff frequency f_c (GHz)	3.750	3.750
Phase velocity v_p (Mm/s)	339.62	339.62

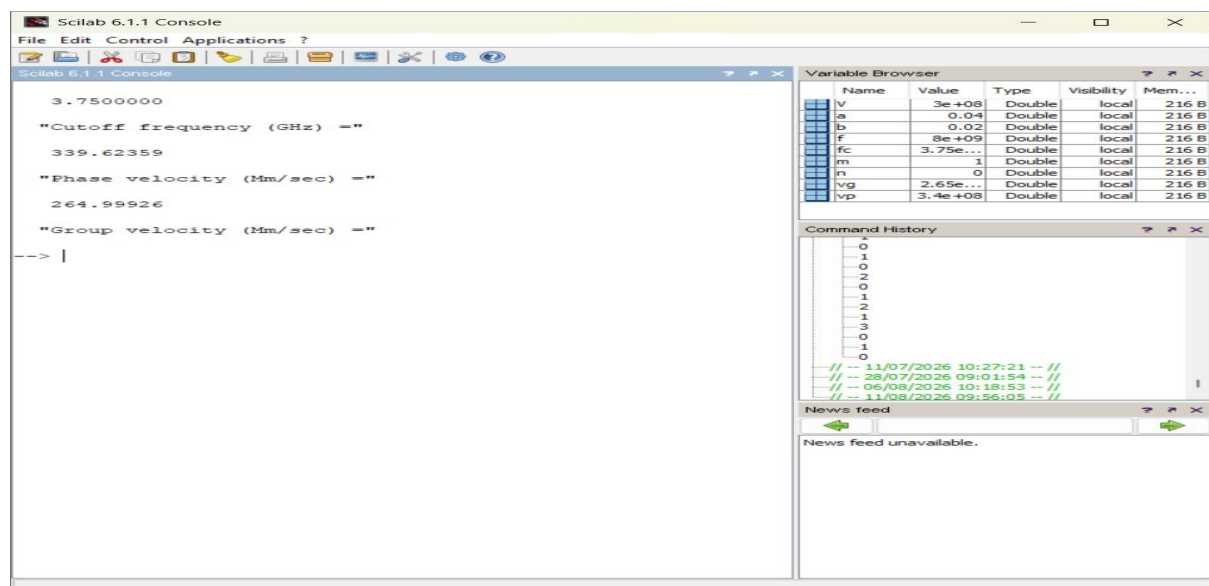
Group velocity v_g (Mm/s)	265.00	265.00
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Reduced width: $a = 3.0$ cm, $b = 2.0$ cm, $f = 8$ GHz, mode TE₁₀

Parameter	Simulation Result	Theoretical Result
Cutoff frequency f_c (GHz)	5.000	5.000
Phase velocity v_p (Mm/s)	384.31	384.31
Group velocity v_g (Mm/s)	234.19	234.19

Table: Variation of Phase and Group Velocity with a ($b = 2.0$ cm, $f = 8$ GHz)

Width a (cm)	f_c (GHz) - Sim	v_p (Mm/s) - Sim	v_g (Mm/s) - Sim	f_c (GHz) - Theory	v_p (Mm/s) - Theory	v_g (Mm/s) - Theory
4.0	3.750	339.62	265.00	3.750	339.62	265.00
3.5	4.286	355.28	253.32	4.286	355.28	253.32
3.0	5.000	384.31	234.19	5.000	384.31	234.19
2.5	6.000	453.56	198.43	6.000	453.56	198.43
2.0	7.500	862.11	104.40	7.500	862.11	104.40



Result:

As the waveguide width a is reduced, the cutoff frequency f_c increases (since $f_c \propto 1/a$). Because the operating frequency stays close to the rising cutoff frequency, the phase velocity v_p increases sharply while the group velocity v_g decreases, showing stronger dispersion as the width approaches the cutoff condition. Reducing a from 4.0 cm to 2.0 cm nearly doubles the cutoff frequency and more than doubles v_p while more than halving v_g .

Test Case 2

A radar system operates at $f = 15$ GHz with waveguide dimensions $a = 2.5$ cm and $b = 1.25$ cm. (1) Compute the cutoff frequency for the TE₁₀ mode. (2) Determine the phase and group velocities for the TE₁₀ mode. (3) Suggest strategies to maintain radar signal fidelity if dispersion is critical.

Code -

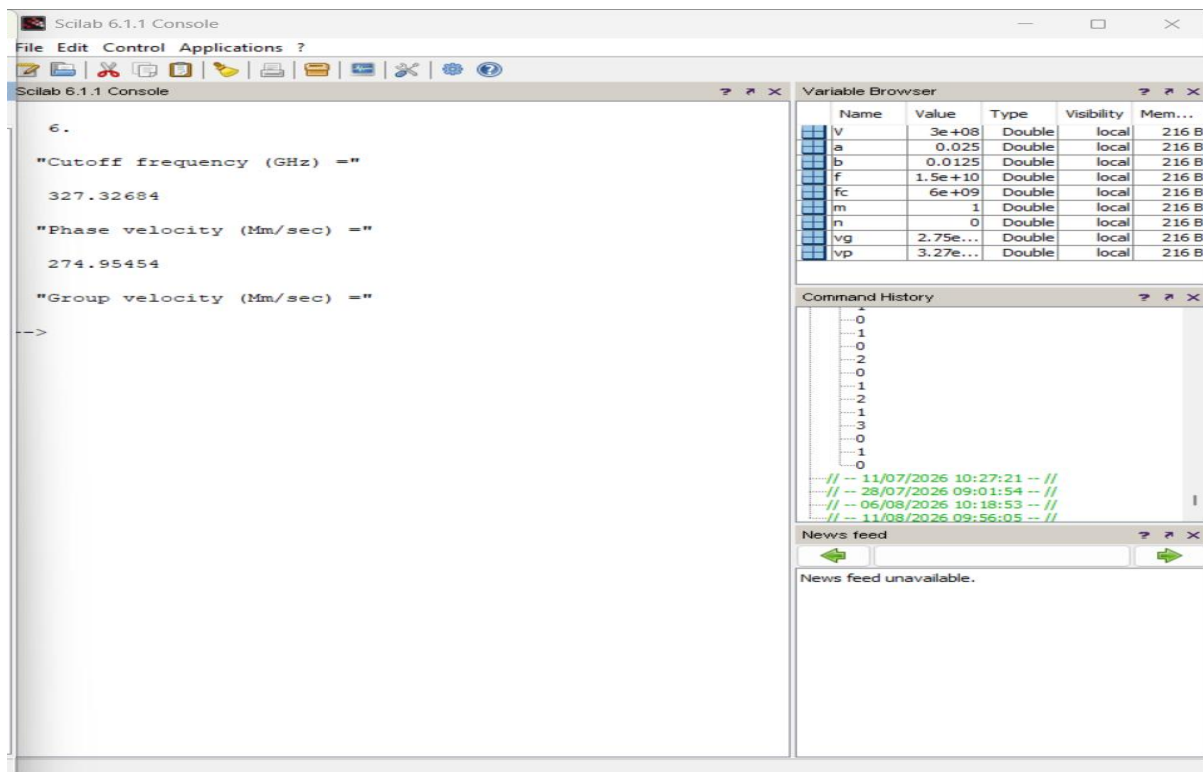
```
clc; clear; close;
V=3*10^8;
a=0.025; b=0.0125; f=15*10^9;
m=1; n=0;
fc=(V/2)*sqrt((m/a)^2+(n/b)^2);
disp(fc*10^-9,'Cutoff frequency (GHz) =');
vp=V/sqrt(1-(fc/f)^2);
disp(vp*10^-6,'Phase velocity (Mm/sec) =');
vg=V*V/vp;
disp(vg*10^-6,'Group velocity (Mm/sec) =');
```

Output:

$a = 2.5$ cm, $b = 1.25$ cm, $f = 15$ GHz, mode TE₁₀

Parameter	Simulation Result	Theoretical Result
Cutoff frequency f_c (GHz)	6.000	6.000

Phase velocity v_p (Mm/s)	327.33	327.33
Group velocity v_g (Mm/s)	274.95	274.95



Result:

For the radar waveguide ($a = 2.5$ cm, $b = 1.25$ cm) operating at 15 GHz, the TE₁₀ cutoff frequency is 6 GHz, well below the operating frequency, giving a moderate dispersion ($v_p = 327.33$ Mm/s, $v_g = 274.95$ Mm/s, both close to c). To preserve signal fidelity where dispersion is critical, the operating frequency should be kept well above cutoff ($f \gg f_c$), a single dominant mode (TE₁₀) should be maintained by suitable waveguide sizing, and dispersion-compensation or narrowband filtering can be used to minimise pulse spreading.

Test Case 3

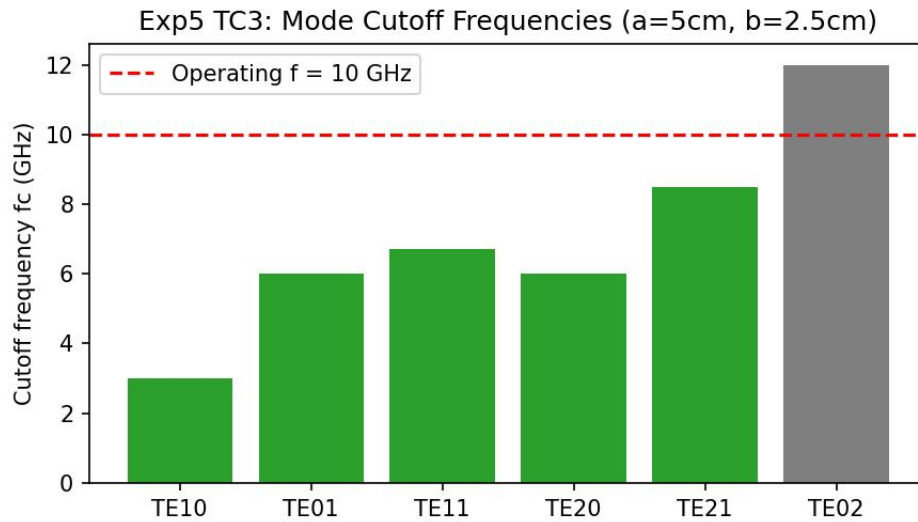
A waveguide with $a = 5.0$ cm and $b = 2.5$ cm carries a signal at $f = 10$ GHz. (1) Determine the number of TE modes that can propagate. (2) For the TE₁₀ mode, calculate the phase and group velocities. (3) Discuss the effect of multiple propagating modes on transmission quality.

Code -

```
clc; clear; close;  
V=3*10^8;  
a=0.05; b=0.025; f=10*10^9;  
modes=[1 0;0 1;1 1;2 0;2 1;0 2];  
for i=1:6  
    m=modes(i,1); n=modes(i,2);  
    fc=(V/2)*sqrt((m/a)^2+(n/b)^2);  
    printf('TE%d%d fc=%f GHz Propagate=%d\n',m,n,fc*10^-9,fc<f);  
end
```

Output:

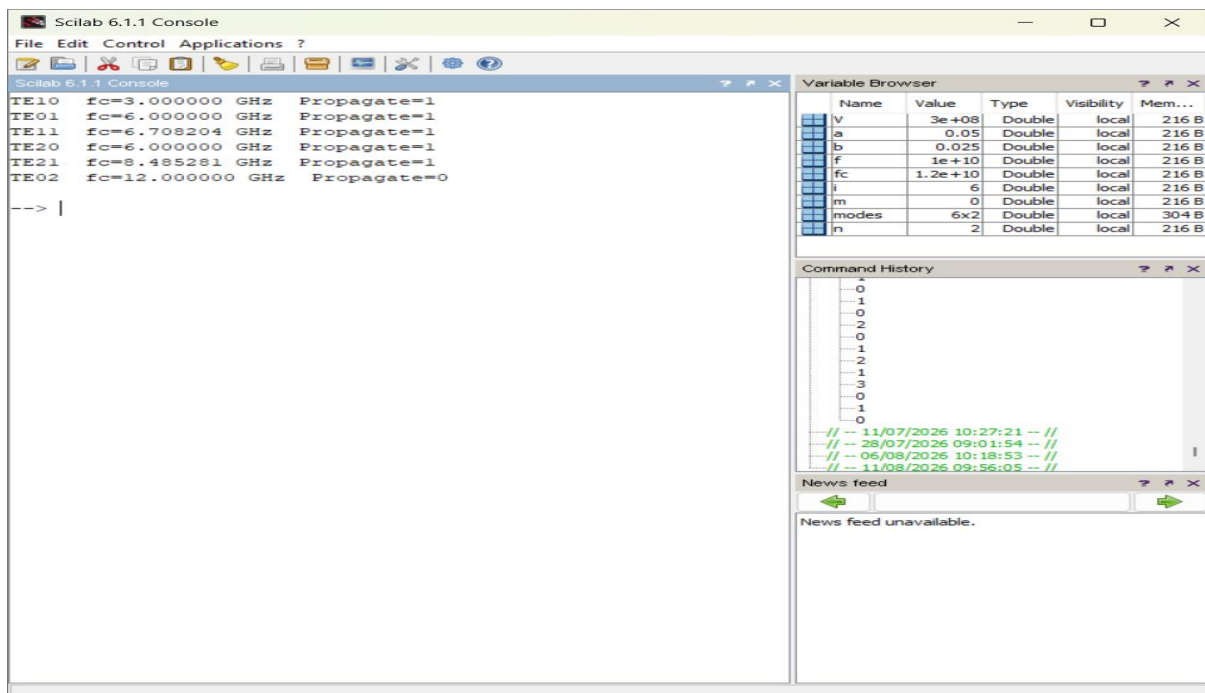
Mode (m,n)	Cutoff Frequency fc (GHz) - Sim	Propagation (Y/N) - Sim	fc (GHz) - Theory	Propagation (Y/N) - Theory
TE10	3.000	Yes	3.000	Yes
TE01	6.000	Yes	6.000	Yes
TE11	6.708	Yes	6.708	Yes
TE20	6.000	Yes	6.000	Yes
TE21	8.485	Yes	8.485	Yes
TE02	12.000	No	12.000	No



Result: Five TE modes can propagate at 10 GHz — TE10, TE01, TE11, TE20, TE21 (TE02 is cut off).

TE10 mode output (a = 5.0 cm, b = 2.5 cm, f = 10 GHz):

Parameter	Simulation Result	Theoretical Result
Cutoff frequency f_c (GHz)	3.000	3.000
Phase velocity v_p (Mm/s)	314.49	314.49
Group velocity v_g (Mm/s)	286.18	286.18



Result:

Five TE modes (TE₁₀, TE₀₁, TE₁₁, TE₂₀, TE₂₁) can propagate in the waveguide ($a = 5.0$ cm, $b = 2.5$ cm) at 10 GHz, while TE₀₂ is cut off. The presence of several simultaneously propagating modes causes each mode to travel with a different group velocity, leading to modal dispersion, signal distortion and inter-mode interference. This can be mitigated in practice by choosing waveguide dimensions such that only the dominant TE₁₀ mode propagates (single-mode operation), using mode filters, or operating well below the cutoff frequency of the next higher-order mode.

Test Case 4

A research lab experiments with high-order TE modes ($m, n > 1$) in a waveguide with $a = 3.0$ cm, $b = 1.5$ cm and $f = 12$ GHz. (1) Calculate the cutoff frequency for the TE₁₀ mode. (2) Compute the phase and group velocities of this mode. (3) Discuss how high-order modes can be used for mode multiplexing.

Code -

```
clc; clear; close;
V=3*10^8;
a=0.03; b=0.015; f=12*10^9;
modes=[1 0;2 0;0 1;1 1;2 1;3 0];
for i=1:6
    m=modes(i,1); n=modes(i,2);
    fc=(V/2)*sqrt((m/a)^2+(n/b)^2);
    printf('TE%d%d fc=%f GHz Propagate=%d\n',m,n,fc*10^-9,fc<f);
end
// TE10 velocities
m=1; n=0;
```

$fc_{10} = (V/2) * \sqrt{(m/a)^2 + (n/b)^2}$;

$vp = V / \sqrt{1 - (fc_{10}/f)^2}$; $vg = V * V / vp$;

$\text{disp}(vp * 10^{-6}, \text{'Phase velocity (Mm/sec) ='})$; $\text{disp}(vg * 10^{-6}, \text{'Group velocity (Mm/sec) ='})$;

Output:

Mode (m,n)	Cutoff Frequency fc (GHz) - Sim	Propagation (Y/N) - Sim	fc (GHz) - Theory	Propagation (Y/N) - Theory
TE10	5.000	Yes	5.000	Yes
TE20	10.000	Yes	10.000	Yes
TE01	10.000	Yes	10.000	Yes
TE11	11.180	Yes	11.180	Yes
TE21	14.142	No	14.142	No
TE30	15.000	No	15.000	No

```

Scilab 6.1.1 Console
File Edit Control Applications ?
330.01146
"Phase velocity (Mm/sec) ="
272.71780
"Group velocity (Mm/sec) ="
--> |

TE10 fc=5.000000 GHz Propagate=1
TE20 fc=10.000000 GHz Propagate=1
TE01 fc=10.000000 GHz Propagate=1
TE11 fc=11.180340 GHz Propagate=1
TE21 fc=14.142136 GHz Propagate=0
TE30 fc=15.000000 GHz Propagate=0

Variable Browser
Name Value Type Visibi... Mem...
V 3e+08 Double local 216 B
a 0.03 Double local 216 B
b 0.015 Double local 216 B
f 1.2e+10 Double local 216 B
fc 1.5e+10 Double local 216 B
fc10 5e+09 Double local 216 B
m 6 Double local 216 B
n 1 Double local 216 B
imodes 6x2 Double local 304 B
in 0 Double local 216 B
Command History
0
1
2
0
0
1
2
1
3
0
1
0
// -- 11/07/2026 10:27:21 -- //
// -- 28/07/2026 09:01:54 -- //
// -- 06/08/2026 10:18:53 -- //
// -- 11/08/2026 09:56:05 -- //
News feed
News feed unavailable.

```

TE10 mode output (a = 3.0 cm, b = 1.5 cm, f = 12 GHz):

Parameter	Simulation Result	Theoretical Result
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Cutoff frequency f_c (GHz)	5.000	5.000
Phase velocity v_p (Mm/s)	330.01	330.01
Group velocity v_g (Mm/s)	272.72	272.72

Result:

At $f = 12$ GHz, four modes (TE₁₀, TE₂₀, TE₀₁, TE₁₁) can propagate in the waveguide ($a = 3.0$ cm, $b = 1.5$ cm), while TE₂₁ and TE₃₀ are cut off. The dominant TE₁₀ mode has $v_p = 330.01$ Mm/s and $v_g = 272.72$ Mm/s. Because several higher-order modes propagate simultaneously in this waveguide, they can, in principle, be exploited for mode-division multiplexing in advanced communication systems — using orthogonal field patterns of different modes to carry independent data channels within the same waveguide, provided mode-selective launching and detection are used to minimise cross-mode coupling.